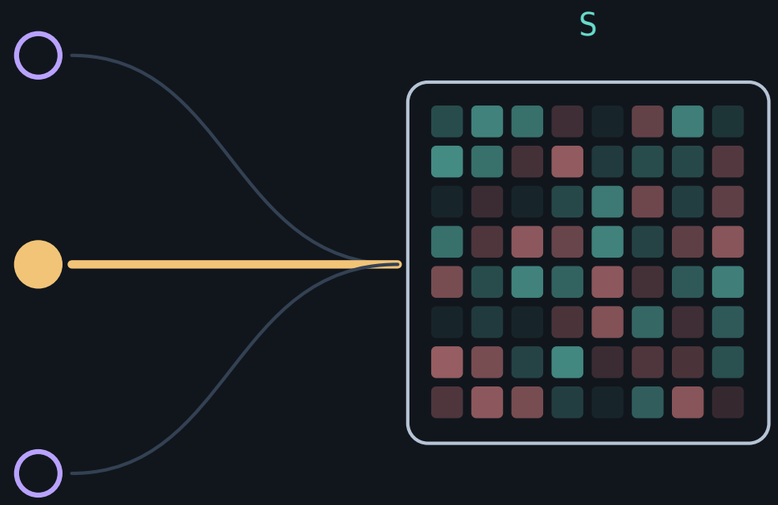
How LLMs recall

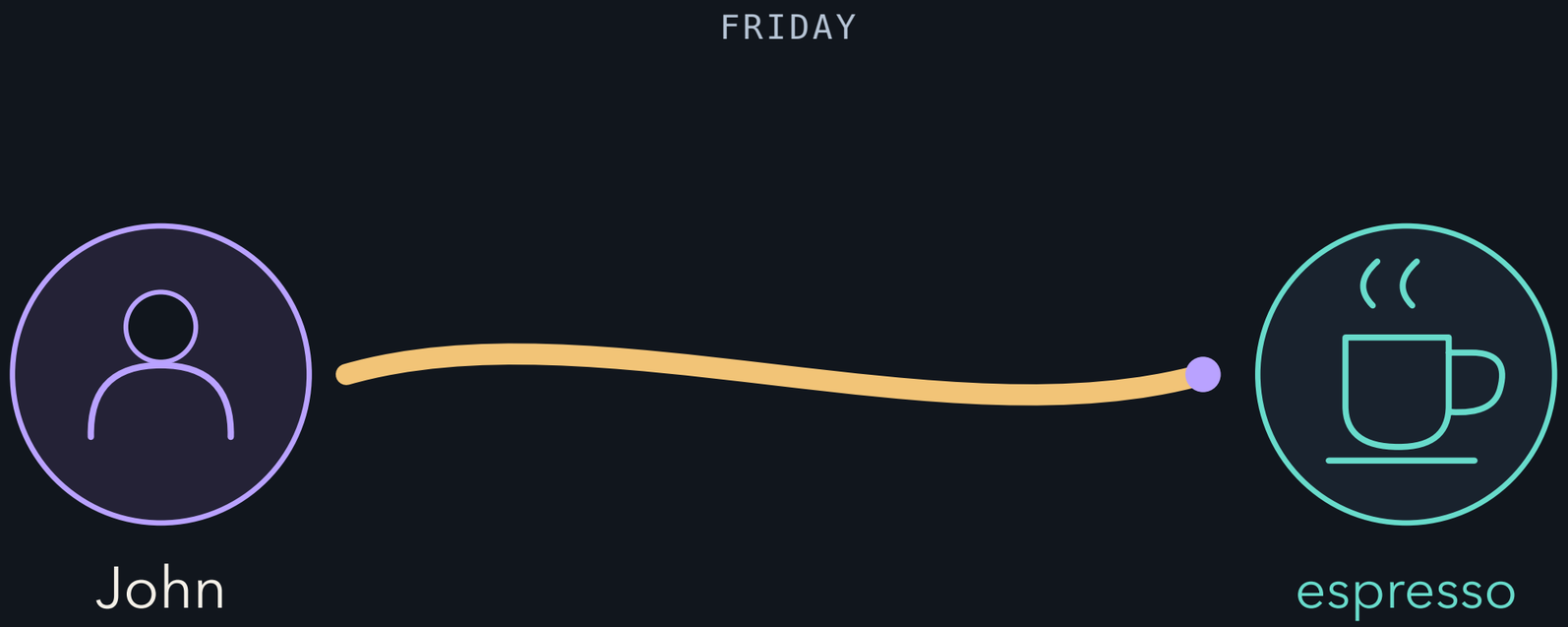
Ayoub Ghriss

ghriss@wustl.edu

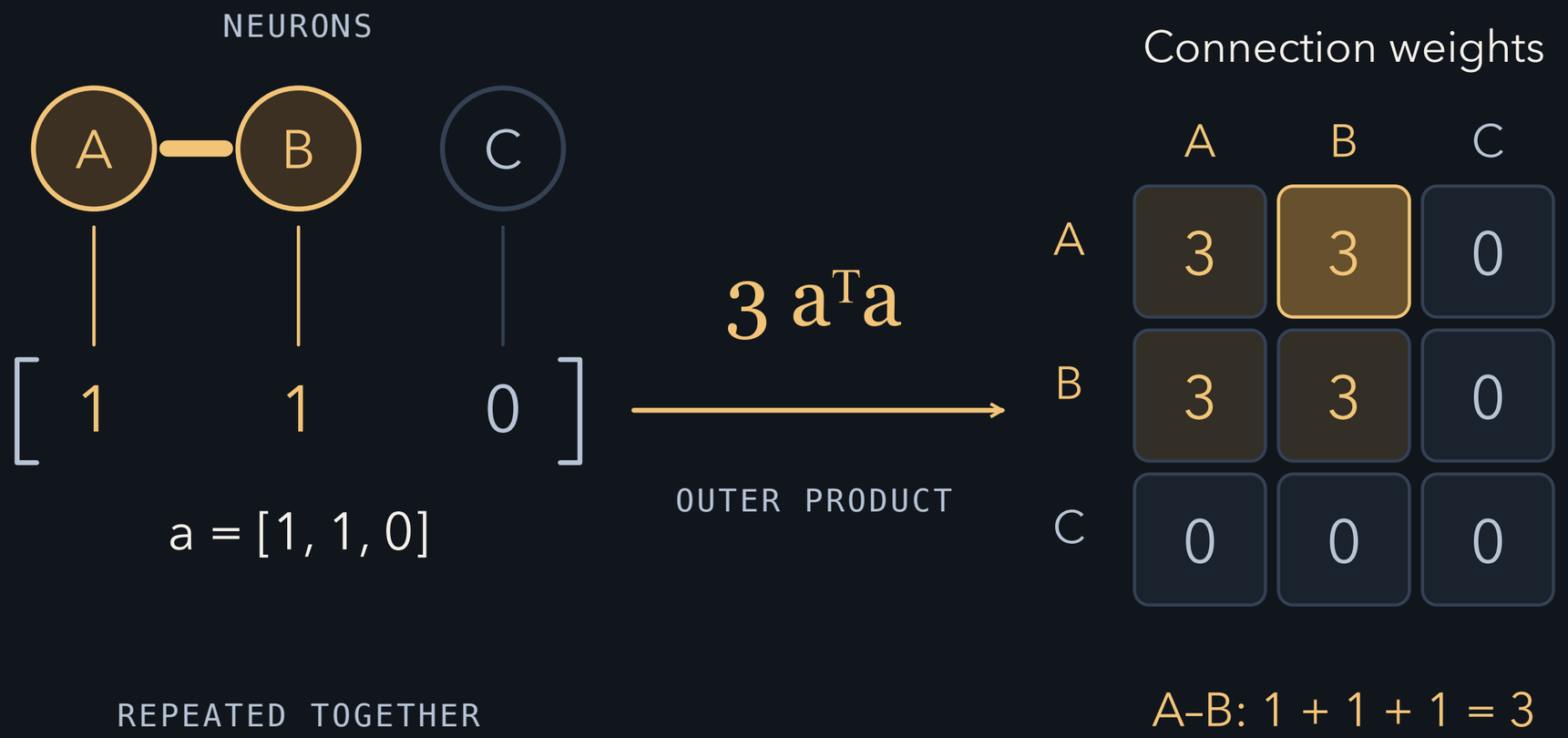
ayghri.me



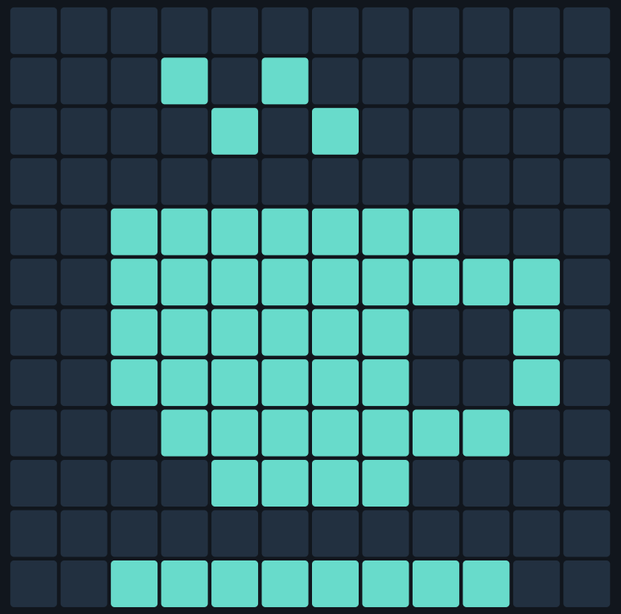
John → ?



An association, in numbers.



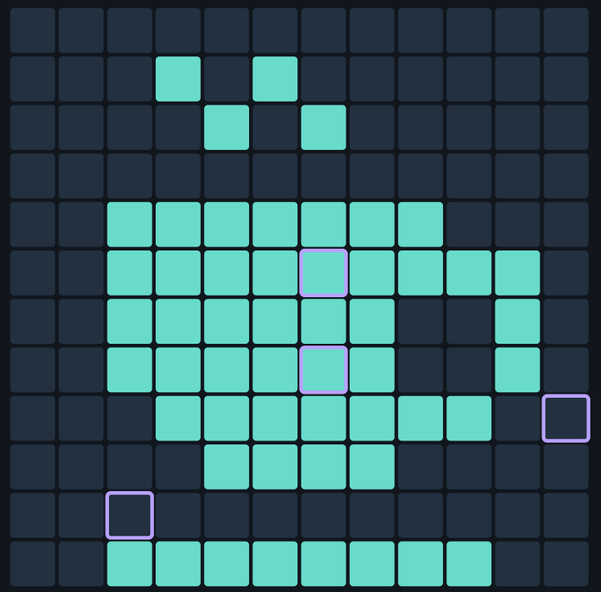
What do you see?



Let the memory settle.

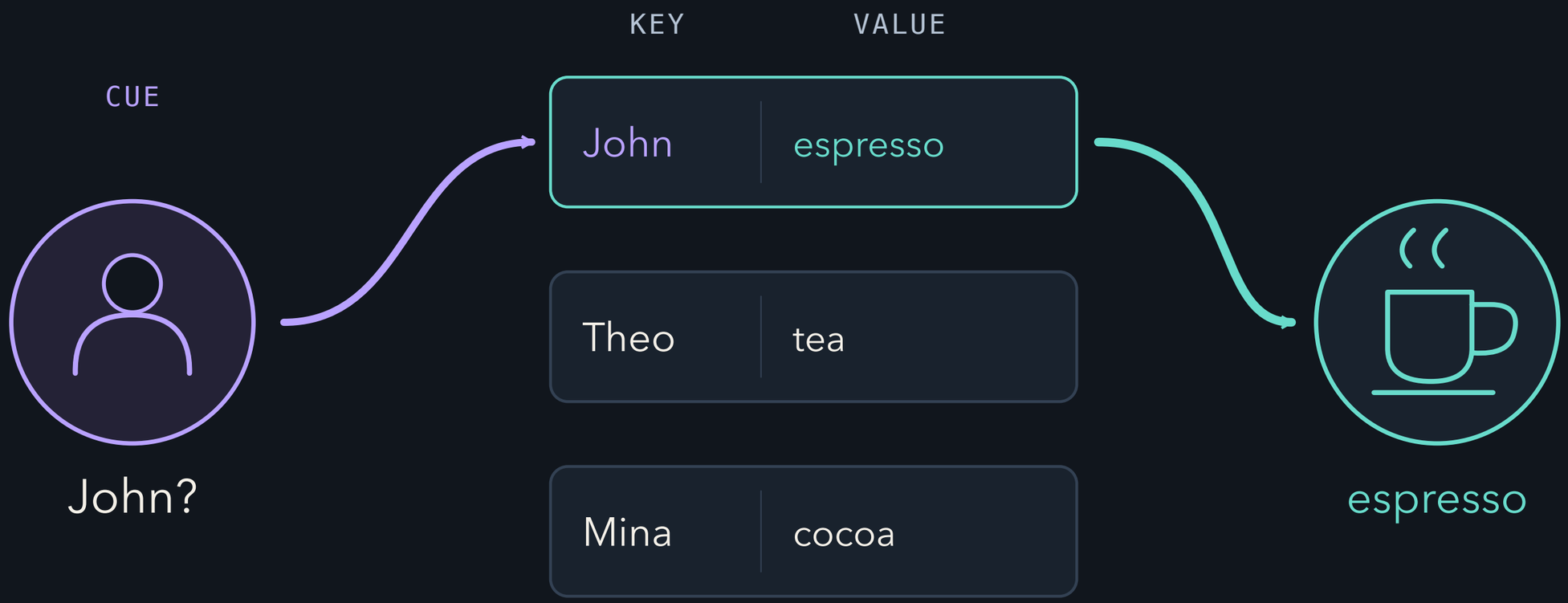
STORED





RECOVERED

Cue → association

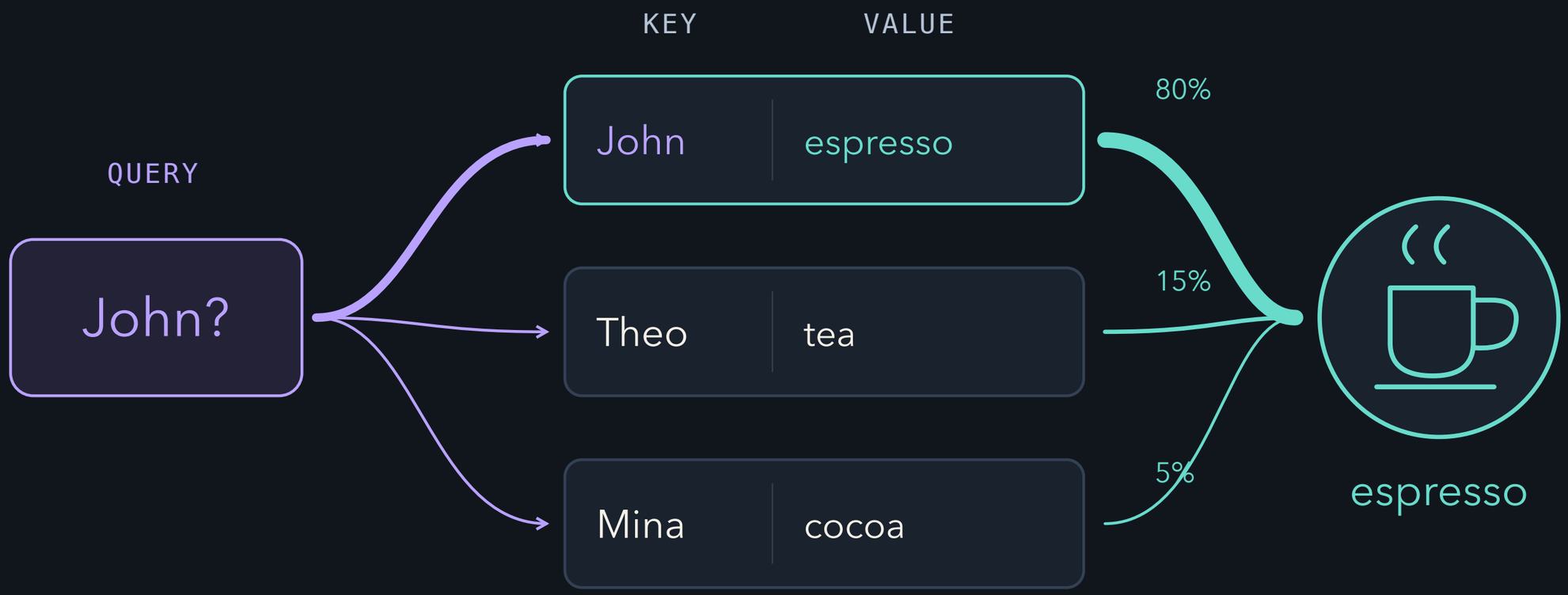


Today’s orders.



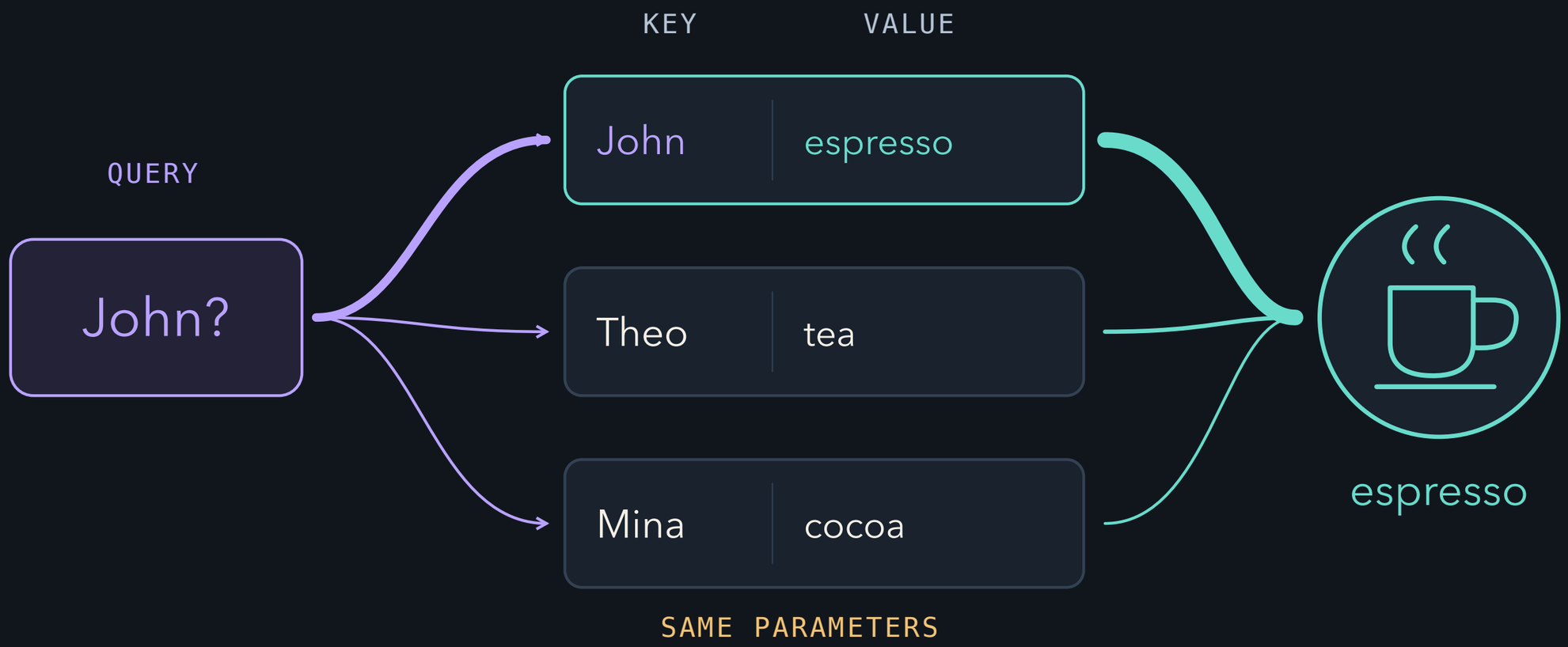
Illustration

What did John order?



Illustration

A new order.

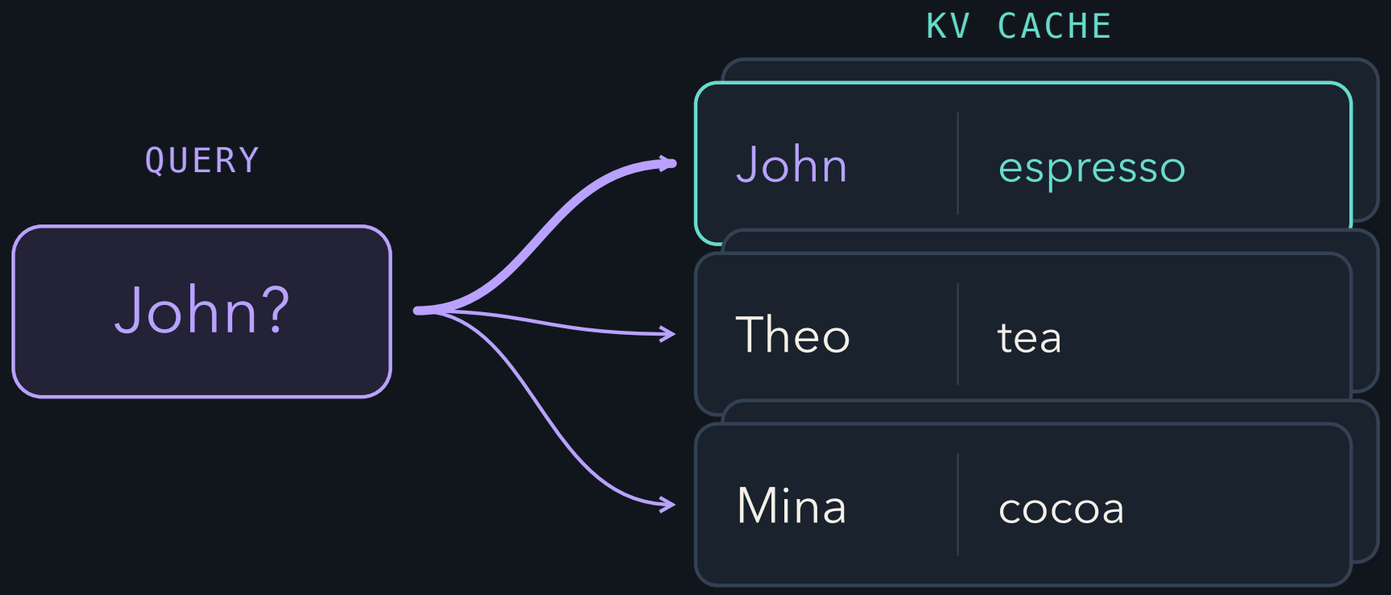


Illustration

Keep the pairs.

$$\sum_i \exp\left(q \cdot k_i\right) v_i$$

SOFTMAX



Separate pairs

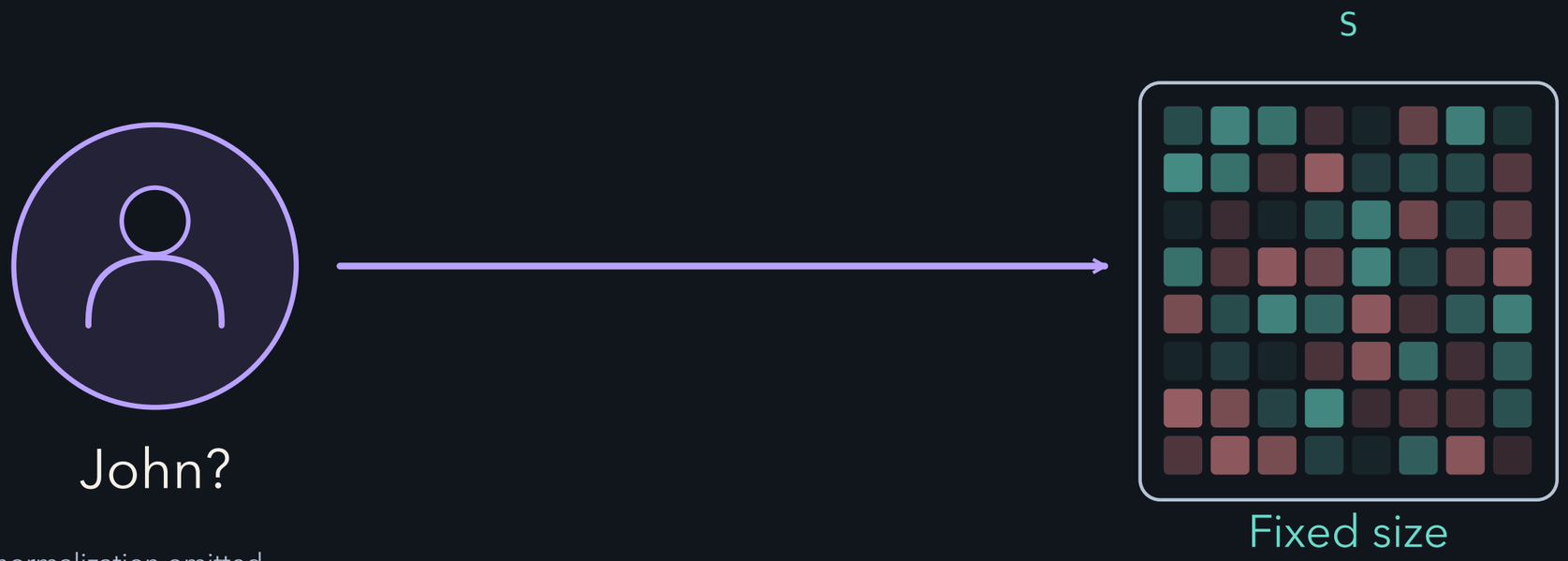
+ more context

Scaling and normalization omitted

Remove exp. Regroup.

q S

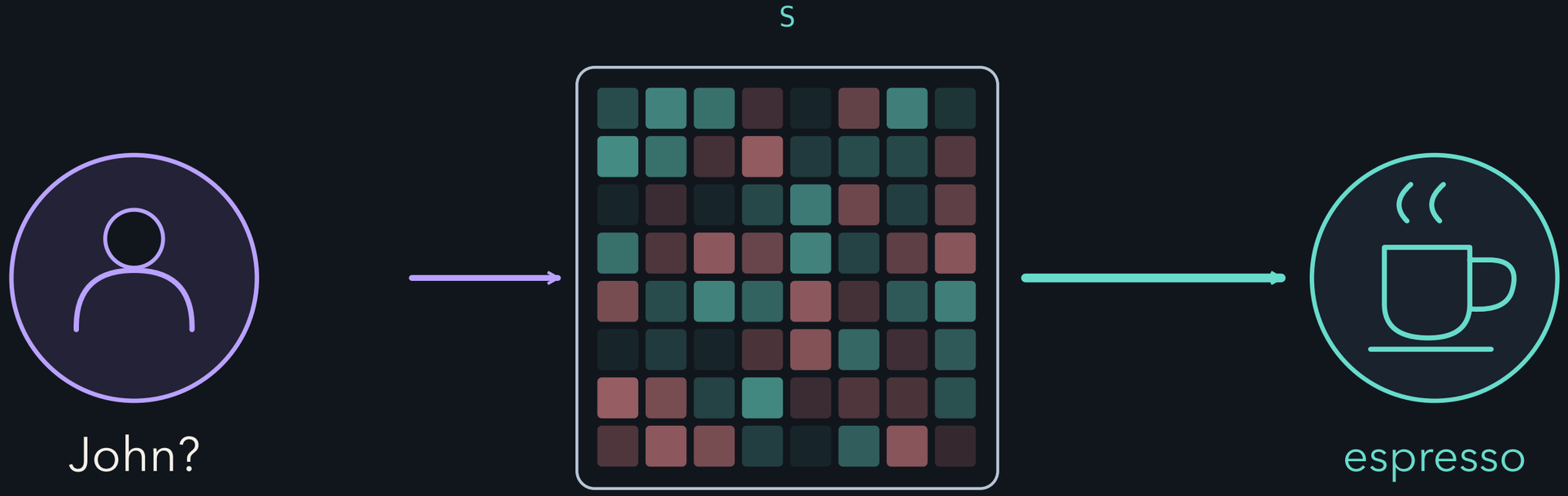
READ



Scaling and normalization omitted

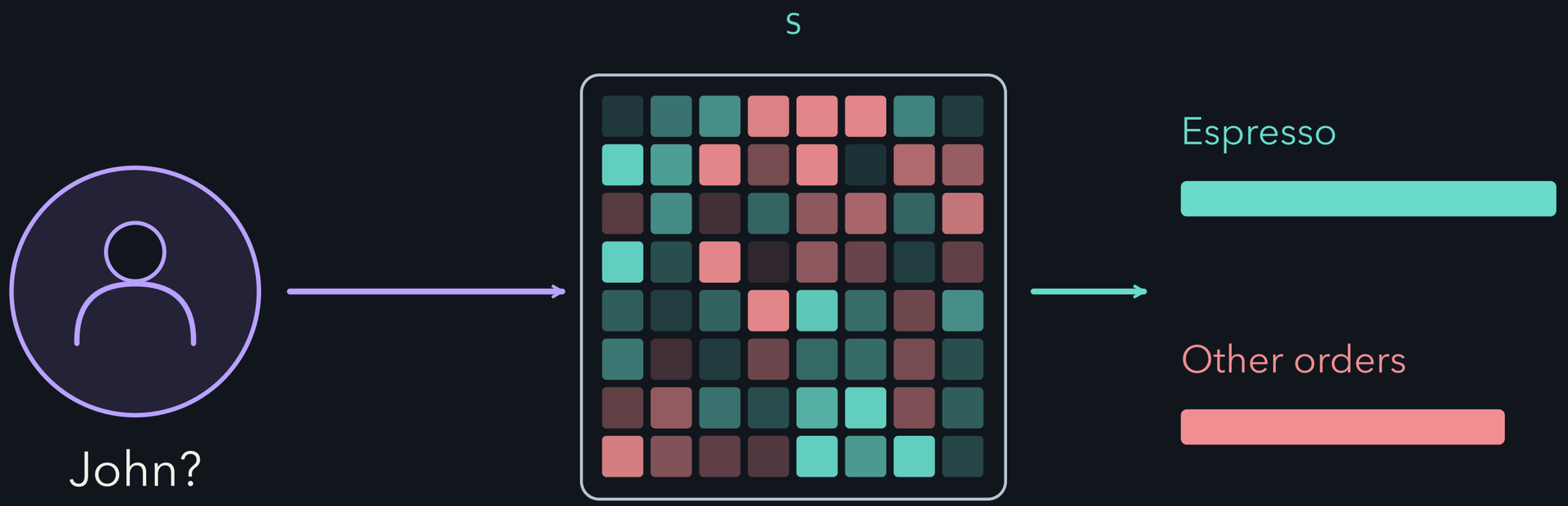
Write. Then read.

$$\text{read} = qS$$



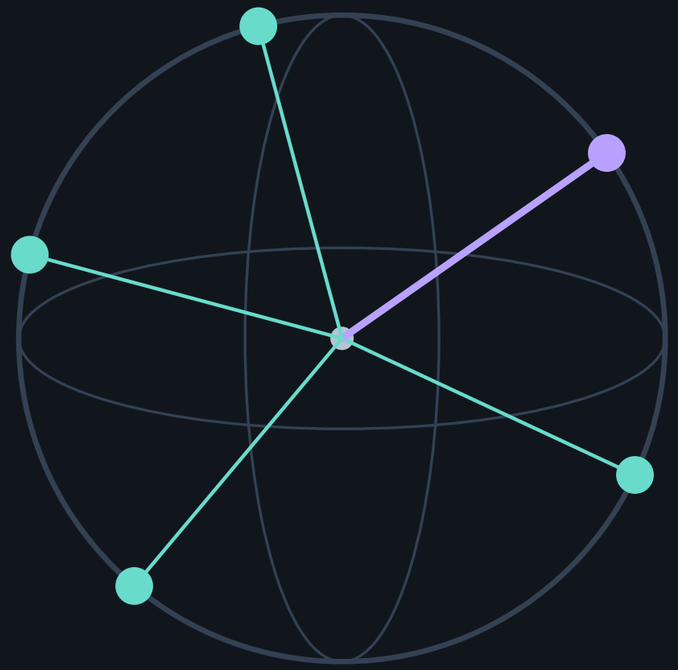
Illustration

How many can we recall?



Illustration

Put the keys on a sphere.

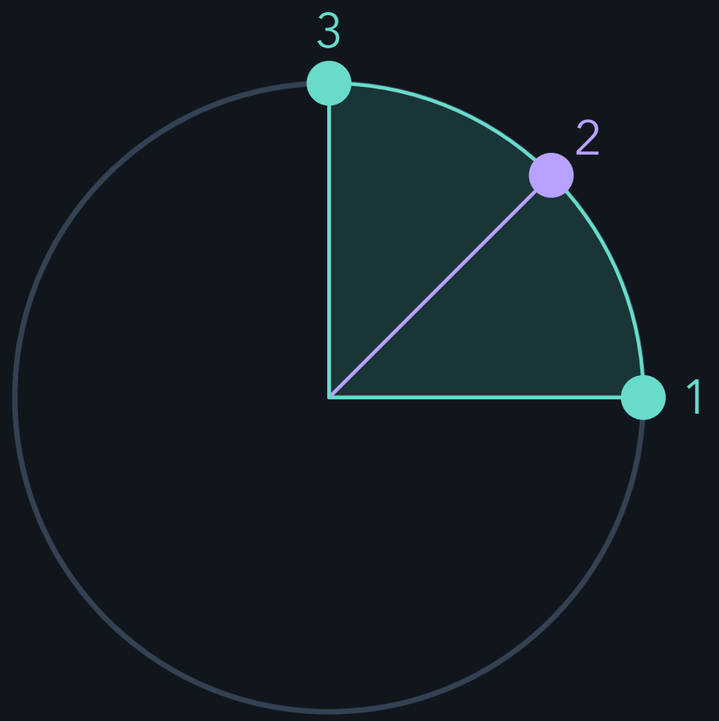


Sphere · great-circle slice

Length = 1

Direction is the address

Spread the addresses.



Spread in sequence

Positional embeddings

Read position 2

Self

1

Largest other match

0.71

Less interference

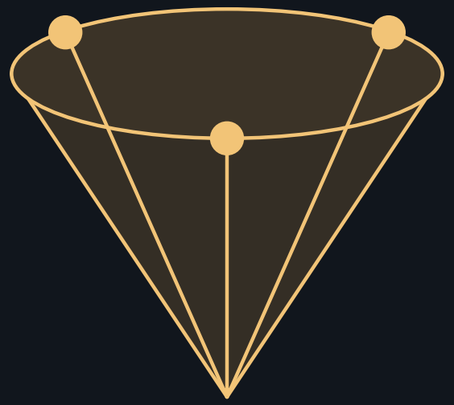
Choose the geometry.

Positive orthant



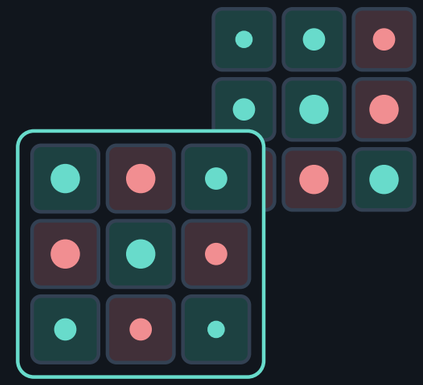
Positive coordinates

Lorentz cone



A constrained cap

PSD matrices



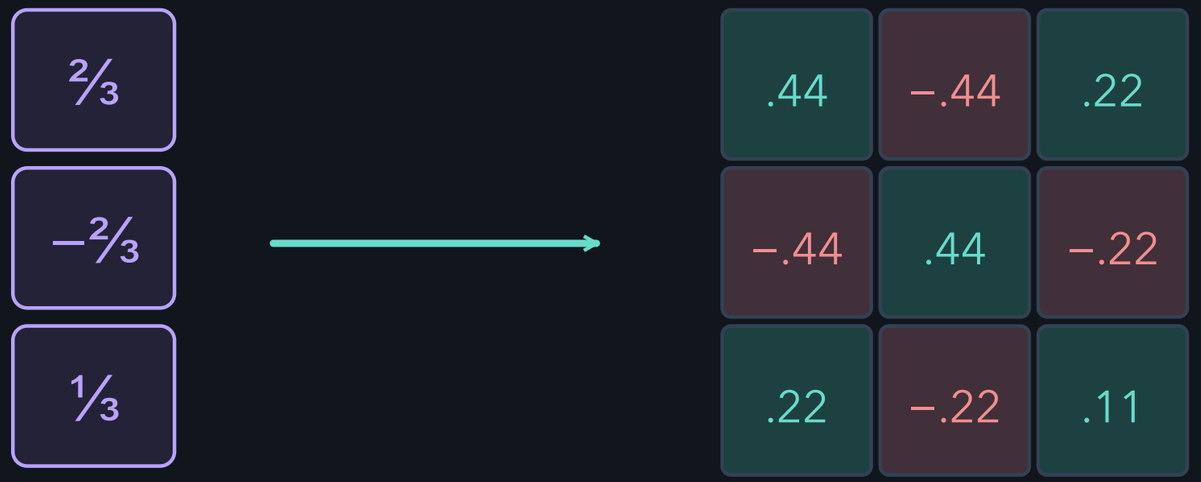
Matrix-valued addresses

Different geometries. Nonnegative matches.

Make the key a matrix.

k

$\mathbf{k}^{\mathrm{T}}\mathbf{k}$



Unit vector

Unit PSD matrix

$$\langle \mathrm{q}, \mathrm{k} \rangle \rightarrow \langle \mathrm{q}, \mathrm{k} \rangle^2$$

$1 \rightarrow 1$

$0.1 \rightarrow 0.01$

Sharper retrieval

PSD: more recall capacity.

IDEALIZED RECALL · SAME PROJECTION WIDTH P

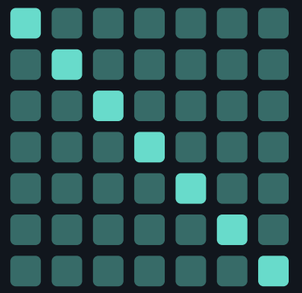
Linear



$\sim p$



PSD



$\sim p^2$

Larger feature state. Still fixed as context grows.

John changes his order.



ADDED ≠ REPLACED

Illustration

Read before writing.

$$\hat{v}_t = k_t \; S_{t-1}$$

NEW

RECALLED



orange juice

espresso

correction

$$e_t = v_t - k_t \; S_{t-1}$$

Illustration

Write the correction.

$$S_t = S_{t-1} + \beta_t \, k_t^{\mathrm{T}} \, e_t$$

S



SAME STATE SIZE

Illustration

Recall. Update.



GEOMETRY → CAPACITY

DELTA → UPDATE

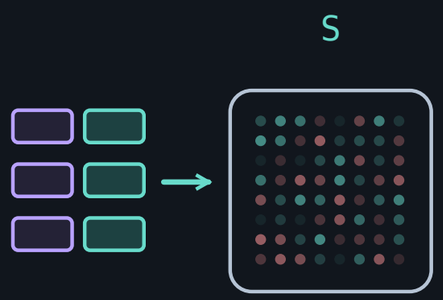
Illustration

Main ideas.



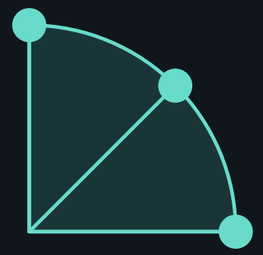
Recall

Match the cue



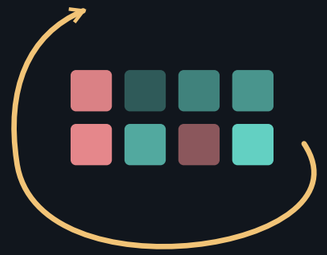
Compress

Pairs → one state



Separate

Geometry sets capacity



Update

Write the error

Open questions.

01 Recall per byte?

02 Order without collisions?

03 What should we forget?

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